

Electrochemistry

THE NERNST EQUATION

We know experimentally that the potential of a single electrode or half-cell varies with the concentration of ions in the cell. In 1889 Walter Nernst derived a mathematical relationship which enable us to calculate the half-cell potential, E , from the standard electrode potential, E° , and the temperature of the cell. This relation known as the Nernst equation can be stated as

$$E = E^\circ - \frac{2.303 RT}{nF} \log K \quad \dots(1)$$

where

E° = standard electrode potential

R = gas constant

T = Kelvin temperature

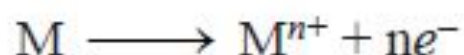
n = number of electrons transferred in the half-reaction

F = Faraday of electricity

K = equilibrium constant for the half-cell reaction as in equilibrium law

Calculation of Half-cell potential

For an oxidation half-cell reaction when the metal electrode M gives M^{n+} ion,



the Nernst equation takes the form

$$E = E^{\circ} - \frac{2.303 RT}{nF} \log \frac{[M^{n+}]}{[M]} \quad \dots(2)$$

The concentration of solid metal [M] is equal to zero. Therefore, the Nernst equation can be written as

$$E = E^{\circ} - \frac{2.303 RT}{nF} \log [M^{n+}] \quad \dots(3)$$

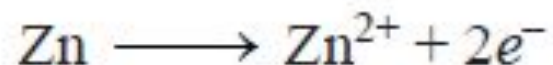
Substituting the values of R, F and T at 25°C , the quantity $2.303 RT/F$ comes to be 0.0591. Thus the Nernst equation (3) can be written in its simplified form as

$$E = E^{\circ} - \frac{0.0591}{n} \log [M^{n+}]$$

This is the equation for a half-cell in which oxidation occurs. In case it is a reduction reaction, the sign of E will have to be reversed.

What is the potential of a half-cell consisting of zinc electrode in 0.01M ZnSO₄ solution at 25°C, $E^\circ = 0.763$ V.

The half-cell reaction is



The Nernst equation for the oxidation half-cell reaction is

$$E = E^\circ - \frac{0.0591}{n} \log[\text{Zn}^{2+}]$$

The number of electrons transferred $n = 2$ and $E^\circ = 0.763$ V.

Substituting these values in the Nernst equation we have

$$\begin{aligned} E &= 0.763 - \frac{0.0591}{2} \log(0.01) \\ &= 0.763 - \frac{0.0591}{2} (-2) \\ &= 0.763 + 0.0591 = \mathbf{0.8221 \text{ V}} \end{aligned}$$

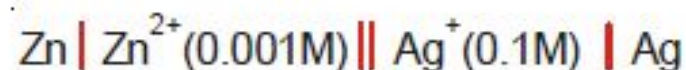
Calculation of Cell potential

The Nernst equation is applicable to cell potentials as well. Thus,

$$E_{\text{cell}} = E^{\circ}_{\text{cell}} - \frac{0.0591}{n} \log K$$

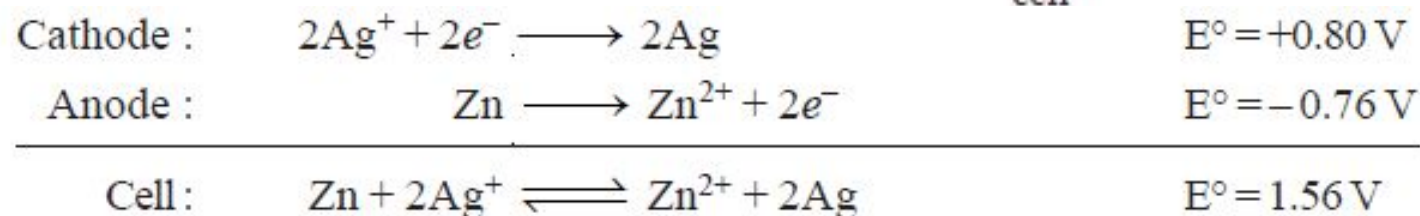
K is the equilibrium constant of the redox cell reaction.

SOLVED PROBLEM. Calculate the emf of the cell.



The standard potential of Ag/Ag^+ half-cell is $+0.80\text{ V}$ and Zn/Zn^{2+} is -0.76 V .

Step 1. Write the half-cell reactions of the anode and the cathode. Then add the anode and cathode half reactions to obtain the cell reaction and the value of E°_{cell} .



Step 2. K for the cell reaction $= \frac{[\text{Zn}^{2+}]}{[\text{Ag}^+]^2}$

substitute the given values in the Nernst equation and solving for E_{cell} , we have

$$\begin{aligned} E_{\text{cell}} &= E^\circ_{\text{cell}} - \frac{0.0591}{n} \log K \\ &= 1.56 - \frac{0.0591}{2} \log \frac{[\text{Zn}^{2+}]}{[\text{Ag}^+]^2} \\ &= 1.56 - \frac{0.0591}{2} \log \frac{[10^{-3}]}{[10^{-1}]^2} \\ &= 1.56 - 0.02955 (\log 10^{-1}) \\ &= 1.56 + 0.02955 \end{aligned}$$

Calculation of Equilibrium constant for the cell reaction

The Nernst equation for a cell is

$$E_{\text{cell}} = E^{\circ}_{\text{cell}} - \frac{0.0591}{n} \log K$$

At equilibrium, the cell reaction is balanced and the potential is zero. The Nernst equation may, now, be written as

$$0 = E^{\circ}_{\text{cell}} - \frac{0.0591}{n} \log K$$

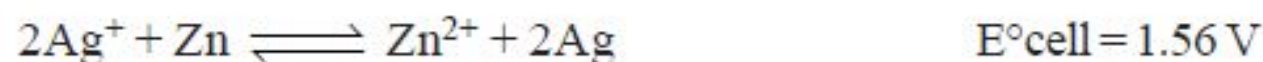
or

$$\log K = \frac{n E^{\circ}_{\text{cell}}}{0.0591}$$

SOLVED PROBLEM. Calculate the equilibrium constant for the reaction between silver nitrate and metallic zinc.

SOLUTION

Step 1. Write the equation for the reaction



Step 2. Substitute values in the Nernst equation at equilibrium

$$\begin{aligned}\log K &= \frac{nE^\circ_{\text{cell}}}{0.0591} \\ 0 &= 1.56 - 0.03 \log K \\ -1.56 &= -0.03 \log K \\ \log K &= \frac{-1.56}{-0.03} = 52 \\ K &= 1 \times 10^{52}\end{aligned}$$

